Krutik Tikam

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GitHub | LinkedIn

Professional Summary

AI and backend engineer building local-first LLM, RAG and computer-vision systems in Python and FastAPI, with measured latency and memory budgets on consumer hardware (RTX 3050, 4 GB VRAM).

Technical Skills

AI/ML: PyTorch, YOLOv8, OpenCV, ByteTrack, Kalman Filtering, RAG (ChromaDB, sentence-transformers), Local LLMs (Ollama), Faster-Whisper, SymPy
Backend: Python, FastAPI (Async), Flask, Node.js, PostgreSQL, SQLite (WAL), REST, WebSockets
Other: Rust (Tauri), React, TypeScript, Git/GitHub Actions, Postman

AI Systems & Backend Projects

Sports Analyser: Broadcast Soccer Tracking & Tactical Analysis [GitHub] | *YOLOv8, PyTorch, OpenCV, MobileNetV3*
- Fine-tuned **YOLOv8** at 1280 px on a custom soccer broadcast dataset (RTX 3050 laptop GPU): **84.4%** mAP@50 on the validation split (**92.0%** on players), **8.3 ms** per-frame detection.
- Built a fused tracker combining IoU, 128-d MobileNetV3 re-ID embeddings (6.2 ms for 22 crops), CIELAB team clustering and affine camera-motion compensation; **0 cross-team ID swaps** on a 120-frame test clip.
- Projected foot-contact points onto a 105 m × 68 m pitch model with homography; fixed a frame-skip time-delta bug that inflated sprint speeds past 65 km/h (now max 36 km/h). End-to-end pipeline runs at 87 ms per frame (about 11.5 FPS).

OmniMath-Local: Local Math & Code Reasoning Backend [GitHub] | *FastAPI, ChromaDB, Ollama, SymPy*
- Built an async **FastAPI** backend serving a local DeepSeek-R1 1.5B model through Ollama with Pydantic-validated JSON outputs, routing each query to a math or code solver.
- Indexed **12,393** text chunks from four textbooks and programming guides in **ChromaDB** (all-MiniLM-L6-v2) for retrieval-augmented answers.
- Added a verification layer: SymPy checks of derivatives, integrals, limits and equations, plus AST (Python), Tree-Sitter (JavaScript) and Ruff checks on generated code.
- Benchmarked on an RTX 3050 (4 GB): about **1.8 GB** total GPU memory in use during inference, **2.06 s** average request latency, about 475 MB on disk.

AURA HUD: Offline Desktop AI Assistant [GitHub] | *Tauri v2 (Rust), React, FastAPI, Faster-Whisper, Ollama*
- Built a hotkey-summoned desktop assistant with a Rust/Tauri shell and a Python FastAPI sidecar; the Rust shell idles at **6.5 MB** RAM (main process only, excluding the WebView and sidecar).
- Streamed raw 16 kHz audio over WebSockets to **INT8 Faster-Whisper** for fully offline voice input; local LLM replies via Ollama in under 3 s.
- Added clipboard-aware actions (Rust arboard) with SQLite WAL history storage.

Other Projects
- **Robotic Arm RL Platform:** custom Gymnasium environment around a 5-DOF Blender arm; trained a PPO policy (Stable-Baselines3, 10,000 timesteps) with a custom reward; headless runs at 700–850 FPS.
- **PokéArchitect:** full-stack team builder and battle analytics app (React, FastAPI, PostgreSQL) deployed on Render, with 30+ REST endpoints.

Professional Experience

Talent Corner HR Services Private Limited — Mumbai, India
Software Engineering Intern — *Dec 2025 – Feb 2026*
- Built Flask REST APIs connecting a Python AI document generator to a React/Node.js frontend.
- Implemented OTP-based login verification.
- Extended **PostgreSQL** schemas with new columns for dashboard user tracking, in a database of **60,000+** professional profiles.
- Load-tested the APIs with Postman.

Education

Nagindas Khandwala College — Mumbai, Maharashtra
B.Sc. Computer Science (Spl. in Artificial Intelligence & Machine Learning) — *2023 – 2026*

Certifications

Cloud Engineering: AWS Data Engineering for Generative AI Applications, AWS Connecting Systems and Machines for Industrial Manufacturing, Microsoft Azure Cloud Computing (SDP)
Machine Learning: DeepLearning.AI Machine Learning Specialization, TensorFlow Developer Certificate